1. For each of the following subsets of $\mathbb{F}^3$, determine whether it is a subspace of $\mathbb{F}^3$.
(a) $\{(x_1, x_2, x_3) \in \mathbb{F}^3 : x_1 + 2x_2 + 3x_3 = 0\}$;
(b) $\{(x_1, x_2, x_3) \in \mathbb{F}^3 : x_1 + 2x_2 + 3x_3 = 4\}$;
(c) $\{(x_1, x_2, x_3) \in \mathbb{F}^3 : x_1 x_2 x_3 = 0\}$;
(d) $\{(x_1, x_2, x_3) \in \mathbb{F}^3 : x_1 = 5x_3\}$.
Solution: (a) $\{(x_1, x_2, x_3) \in \mathbb{F}^3 : x_1 + 2x_2 + 3x_3 = 0\}$ is a subspace of $\mathbb{F}^3$. By 1.34, to show a subset is a subspace, we just need to check Additive identity, Closed under addition and Closed under scalar multiplication.
Additive identity: it is clear that the additive identity $(0, 0, 0)$ of $\mathbb{F}^3$ is contained in $\{(x_1, x_2, x_3) \in \mathbb{F}^3 : x_1 + 2x_2 + 3x_3 = 0\}$.
Closed under addition: if $(a_1, a_2, a_3), (b_1, b_2, b_3) \in \{(x_1, x_2, x_3) \in \mathbb{F}^3 : x_1 + 2x_2 + 3x_3 = 0\}$, the

$$a_1 + 2a_2 + 3a_3 = 0 \quad \text{and} \quad b_1 + 2b_2 + 3b_3 = 0.$$

Hence

$$(a_1 + b_1) + 2(a_2 + b_2) + 3(a_3 + b_3) = (a_1 + 2a_2 + 3a_3) + (b_1 + 2b_2 + 3b_3) = 0,$$

this means $(a_1 + b_1, a_2 + b_2, a_3 + b_3) = (a_1, a_2, a_3) + (b_1, b_2, b_3)$ is also in $\{(x_1, x_2, x_3) \in \mathbb{F}^3 : x_1 + 2x_2 + 3x_3 = 0\}$.
Closed under scalar multiplication: if $(a_1, a_2, a_3) \in \{(x_1, x_2, x_3) \in \mathbb{F}^3 : x_1 + 2x_2 + 3x_3 = 0\}$, then $a_1 + 2a_2 + 3a_3 = 0$. For any $\lambda \in \mathbb{F}$, we have

$$\lambda a_1 + 2(\lambda a_2) + 3(\lambda a_3) = \lambda(a_1 + 2a_2 + 3a_3) = 0.$$

This means

$$\lambda(a_1, a_2, a_3) = (\lambda a_1, \lambda a_2, \lambda a_3) \in \{(x_1, x_2, x_3) \in \mathbb{F}^3 : x_1 + 2x_2 + 3x_3 = 0\}.$$

(b) $\{(x_1, x_2, x_3) \in \mathbb{F}^3 : x_1 + 2x_2 + 3x_3 = 4\}$ is not a subspace of $\mathbb{F}^3$ since $(0, 0, 0)$ is not in it.
(c) $\{(x_1, x_2, x_3) \in \mathbb{F}^3 : x_1 x_2 x_3 = 0\}$ is not a subspace of $\mathbb{F}^3$ since $(1, 1, 0)$ and $(0, 1, 1)$ are in it, but the sum $(1, 2, 1) = (1, 1, 0) + (0, 1, 1)$ is not in it.
(d) $\{(x_1, x_2, x_3) \in \mathbb{F}^3 : x_1 = 5x_3\}$ is a subspace of $\mathbb{F}^3$.
Additive identity: it is clear that the additive identity $(0, 0, 0)$ of $\mathbb{F}^3$ is contained in $\{(x_1, x_2, x_3) \in \mathbb{F}^3 : x_1 = 5x_3\}$.
Closed under addition: if $(a_1, a_2, a_3), (b_1, b_2, b_3) \in \{(x_1, x_2, x_3) \in \mathbb{F}^3 : x_1 = 5x_3\}$, then $a_1 = 5a_3$ and $b_1 = 5b_3$. Hence

$$a_1 + b_1 = 5a_3 + 5b_3 = 5(a_3 + b_3),$$

this means $(a_1 + b_1, a_2 + b_2, a_3 + b_3) = (a_1, a_2, a_3) + (b_1, b_2, b_3)$ is also in $\{(x_1, x_2, x_3) \in \mathbb{F}^3 : x_1 = 5x_3\}$.
Closed under scalar multiplication: if $(a_1, a_2, a_3) \in \{(x_1, x_2, x_3) \in \mathbb{F}^3 : x_1 = 5x_3\}$, then $a_1 = 5a_3$. For any $\lambda \in \mathbb{F}$, we have $\lambda a_1 = \lambda(5a_3) = 5(\lambda a_3)$. This means

$$\lambda(a_1, a_2, a_3) = (\lambda a_1, \lambda a_2, \lambda a_3) \in \{(x_1, x_2, x_3) \in \mathbb{F}^3 : x_1 = 5x_3\}.$$

2. Verify all the assertions in Example 1.35.
Solution: (a) If this set is a subspace of $\mathbb{F}^4$, then $(0, 0, 0, 0) \in \mathbb{F}^4$, then $0 = 5 \cdot 0 + b$. Hence $b = 0$. Follow the same steps of Problem 1 (d), we will see the set is a subspace of $\mathbb{F}^4$ if $b = 0$.
(b) (c) and (d) is similar to Problem 3 and 4.

Now let us consider (e). Denote the set of all sequences of complex numbers with limit 0 by $A$.
Additive identity: it is clear that $(0, 0, \cdots) \in A$.
Closed under addition: if $(a_1, a_2, \cdots), (b_1, b_2, \cdots) \in A$, then

$$\lim_{n\to\infty} a_n = 0 \quad \text{and} \quad \lim_{n\to\infty} b_n = 0.$$

It is easy to see

$$\lim_{n\to\infty} (a_n + b_n) = \lim_{n\to\infty} a_n + \lim_{n\to\infty} b_n = 0 + 0 = 0.$$

This means $(a_1 + b_1, a_2 + b_2, \cdots) = (a_1, a_2, \cdots) + (b_1, b_2, \cdots) \in A$.
Closed under scalar multiplication: if $(a_1, a_2, \cdots) \in A$, then

$$\lim_{n\to\infty} a_n = 0.$$

For any $\lambda \in \mathbb{C}$, it is easy to see

$$\lim_{n\to\infty} (\lambda a_n) = \lambda \lim_{n\to\infty} a_n = \lambda 0 = 0.$$

This means $\lambda(a_1, a_2, \cdots) = (\lambda a_1, \lambda a_2, \cdots) \in A$.

5. Is $\mathbb{R}^2$ a subspace of the complex vector space $\mathbb{C}^2$?
Solution: Note that we consider complex vector space, so if $\mathbb{R}^2$ is a subspace of the complex vector space $\mathbb{C}^2$, then

$$i(1, 1) = (i, i) \in \mathbb{R}^2,$$

we get a contradiction. Hence $\mathbb{R}^2$ is not a subspace of the complex vector space $\mathbb{C}^2$.
6. (a) Is $\{(a, b, c) \in \mathbb{R}^3 : a^3 = b^3\}$ a subspace of $\mathbb{R}^3$?
(b) Is $\{(a, b, c) \in \mathbb{C}^3 : a^3 = b^3\}$ a subspace of $\mathbb{C}^3$?
Solution: (a) Because $a^3 = b^3$ if and only if $a = b$ in $\mathbb{R}$, hence

$$\{(a, b, c) \in \mathbb{R}^3 : a^3 = b^3\} = \{(a, b, c) \in \mathbb{R}^3 : a = b\}$$

is obviously a subspace of $\mathbb{R}^3$ by the similar arguments in Problem 1 and Problem 2.

7. Give an example of a nonempty subset $U$ of $\mathbb{R}^2$ such that $U$ is closed under addition and under taking additive inverses (meaning $-u \in U$ whenever $u \in U$), but $U$ is not a subspace of $\mathbb{R}^2$.

Solution: Denote $\{(x, y) \in \mathbb{R}^2 : x, y \in \mathbb{Z}\}$ by $U$, then $U$ is not empty. If $(x_1, y_1) \in A$ and $(x_2, y_2) \in A$, then $x_1, x_2, y_1, y_2 \in \mathbb{Z}$. Hence $x_1 + x_2$ and $y_1 + y_2$ a integers. This means $(x_1 + x_2, y_1 + y_2) = (x_1, y_1) + (x_2, y_2) \in U$, i.e. $U$ is closed under addition. Similarly, since $(-x_1, -y_1) \in U$, it follows that $U$ is closed under additive inverses. However, $U$ is not closed under scalar multiplication since $(1, 1) \in A$ while $\frac{1}{2}(1, 1) \notin U$. Hence $U$ is not a subspace of $\mathbb{R}^2$.

8. Give an example of a nonempty subset $U$ of $\mathbb{R}^2$ such that $U$ is closed under scalar multiplication, but $U$ is not a subspace of $\mathbb{R}^2$.

Solution: Denote $\{(x, y) \in \mathbb{R}^2 : x = 0 \text{ or } y = 0\}$ by $U$, then $U$ is not empty. If $(x, 0) \in U$, then for any $\lambda \in \mathbb{R}$, we have

$$\lambda(x, 0) = (\lambda x, 0) \in U.$$

Similarly, $\lambda(0, y) \in U$, hence $U$ is closed under scalar multiplication. However, $(1, 0), (0, 1) \in U$ while $(1, 1) = (1, 0) + (0, 1) \notin U$. This implies $U$ is not closed under addition, hence not a subspace of $\mathbb{R}^2$.

9. A function $f : \mathbb{R} \to \mathbb{R}$ is called periodic if there exists a positive number $p$ such that $f(x) = f(x + p)$ for all $x \in \mathbb{R}$. Is the set of periodic functions from $\mathbb{R}$ to $\mathbb{R}$ a subspace of $\mathbb{R}^{\mathbb{R}}$? Explain.

Solution: Denote the set of periodic functions from $\mathbb{R}$ to $\mathbb{R}$ by $S$. Then $S$ is not a subspace of $\mathbb{R}^{\mathbb{R}}$. Otherwise, we have $h(x) = \sin\sqrt{2}x + \cos x$ since both $f(x) = \sin\sqrt{2}x$ and $g(x) = \cos x$ are periodic functions from $\mathbb{R}$ to $\mathbb{R}$. Assume there exists a positive number $p$ such that $h(x) = h(x + p)$ for all $x \in \mathbb{R}$, then $1 = h(0) = h(p) = h(-p)$. It is equivalent to

$$1 = \cos p + \sin\sqrt{2}p = \cos p - \sin\sqrt{2}p,$$

this implies $\sin\sqrt{2}p = 0$ and $\cos p = 1$. $\cos p = 1$ deduces that $p = 2k\pi$ where $k \in \mathbb{Z}$. However $\sin\sqrt{2}p = 0$ implies $\sqrt{2}p = 2l\pi$ where $l \in \mathbb{Z}$. Hence

10. Suppose $U_1$ and $U_2$ are subspaces of $V$. Prove that the intersection $U_1 \cap U_2$ is a subspace of $V$.

Solution: Additive identity: by definition $0 \in U_1$ and $0 \in U_2$, hence $0 \in U_1 \cap U_2$.

Closed under addition: if $x \in U_1 \cap U_2$ and $y \in U_1 \cap U_2$, then $x \in U_1$ and $y \in U_1$, hence $x + y \in U_1$ for $U_1$ is closed under addition. Similarly, $x + y \in U_2$. Therefore $x + y \in U_1 \cap U_2$.

Closed under scalar multiplication: if $x \in U_1 \cap U_2$, then $x \in U_1$. Then for any $\lambda \in \mathbb{F}$, we have $\lambda x \in U_1$ since $U_1$ is closed under scalar multiplication. Similarly, $\lambda x \in U_2$. Therefore $\lambda x \in U_1 \cap U_2$.

11. Prove that the intersection of every collection of subspaces of $V$ is a subspace of $V$.

Solution: Assume $U_i$ are subspaces of $V$, where $i \in I$. Now we will show $\cap_{i \in I} U_i$ is a subspace of $V$

Additive identity: by definition $0 \in U_i$ for every $i \in I$, hence $0 \in \cap_{i \in I} U_i$.

Closed under addition: if $x \in \cap_{i \in I} U_i$ and $y \in \cap_{i \in I} U_i$, then for any given $i \in I$, we have $x \in U_i$ and $y \in U_i$, hence $x + y \in U_i$ for $U_i$ is closed under addition. Therefore $x + y \in \cap_{i \in I} U_i$.

Closed under scalar multiplication: if $x \in \cap_{i \in I} U_i$, then $x \in U_i$ for any given $i \in I$. Then for any $\lambda \in \mathbb{F}$, we have $\lambda x \in U_i$ since $U_i$ is closed under scalar multiplication. Therefore $\lambda x \in \cap_{i \in I} U_i$.

12. Prove that the union of two subspaces of $V$ is a subspace of $V$ if and only if one of the subspaces is contained in the other.

Solution: Suppose $U$ and $W$ are two subspaces of $V$. We argue it by contradiction. If $U \cup W$ is a subspace of $V$, moreover *[Math Processing Error]* and $W \nsubseteq U$. Consider $u \in U \setminus W$ and $w \in W \setminus U$, then $u + w \in U \cup W$ since $U \cup W$ is a subspace of $V$. Hence $u + w \in U$ or $W$. If $u + w \in U$, then $w = (u + w) - u \in U$. We get a contradiction. If $u + w \in W$, then $u = (u + w) - w \in W$. We also get a contradiction. Hence if $U \cup W$ is a subspace of $V$, we must have $U \subset W$ or $W \subset U$.

If $U \subset W$ or $W \subset U$. Without loss of generality, we can assume $U \subset W$. Then $U \cup W = W$ is obviously a subspace of $V$.

13. Prove that the union of three subspaces of $V$ is a subspace of $V$ if and only if one of the subspaces contains the other two. [\emph{This exercise is surprisingly harder than the previous exercise, possibly because this exercise is not true if we replace F with a field containing only two elements}.]

14. Verify the assertion in Example 1.38